Ensemble learning

Research Project

2026/2027

Stanisław Kaźmierczak

Timetable

1	6.10.2026	Introduction	
2	13.10.2026	**Project 1: Initial plan**	
3	20.10.2026	Project 1: Final plan	
4	27.10.2026	Project consultations	
5	3.11.2026	Project consultations	
6	10.11.2026	**Project 1: Progress checkpoint**	
7	17.11.2026	Project consultations	
8	24.11.2026	**Project 1: Presentation**	
9	1.12.2026	**Project 2: Initial plan**	Project 1: Continued work
10	8.12.2026	Project 2: Final plan	Project consultations
11	15.12.2026	Project consultations	**Project 1: Progress checkpoint**
12	22.12.2026	**Project 2: Progress checkpoint**	Project consultations
13	12.01.2027	Project consultations	**Project 1: Progress checkpoint**
14	19.01.2027	**Project 2: Presentation**	**Article draft based on Project 1**
15	26.01.2027	Project consultations	

Project organisation and assessment

- Work in pairs.
- Complete two projects, worth 50 points each (100 points in total).
- Each project is assessed as follows:
  - Project plan: 10 points.
  - Report, source code, and presentation: 40 points in total.
- Key assessment criteria include:
  - The research process.
  - Report quality.
  - The novelty and originality of the solution.
- If the results of Project 1 are promising, further work aimed at preparing a research paper may replace Project 2.
- In this case, assessment criteria and point allocation are determined individually.

Submission and presentation requirements

- Upload the final project plan to the LeON platform before the start of the session specified in the schedule.
- Upload all required project materials—report, source code, and presentation slides—to LeON before the start of the session in which the project is presented.
- Deliver an approximately 10-minute slide presentation to the instructor.

Late submission and presentation penalties

- Each commenced week of delay counts as a full week.
- Second-week deductions are additional to first-week deductions.

Delayed item	**First week**	**Second week**	**More than two weeks**
Project submission	−5 points	Additional −10 points (−15 total)	0 points for the project
Project plan submission	−2 points	Additional −3 points (−5 total)	0 points for the project plan
Project presentation	−2 points	Additional −4 points (−6 total)	0 points for the project

Attendance and consultations

- Mandatory sessions are marked in bold in the schedule.
- Each absence from a mandatory session results in a 2-point deduction.
- Non-mandatory sessions provide an opportunity for project consultations.
- If you plan to attend a non-mandatory session, notify the instructor by 6 p.m. on the preceding day.

Use of AI, external code, reproducibility

- Clearly disclose the purpose and extent of AI use, specifying which parts of the project plan, report, or code were supported by AI.
- Code from external sources, such as books, articles, or blogs, may be used provided that:
  - The source is cited in the report.
  - The original solution is modified.
- Failure to meet these requirements, or any other form of plagiarism, results in a failing grade.
- Support reproducibility by initializing random number generators with fixed seeds.

Grading scale

Total points	Grade
0–50	2.0
51–60	3.0
61–70	3.5
71–80	4.0
81–90	4.5
91–100	5.0

Project plan

- Present a complete project plan during the Initial plan session.
- Revise the plan based on the instructor's feedback and upload the final version to LeON one week later.
- The plan must be well thought out, detailed, and sufficiently specific to guide implementation.
- Include:
  - A literature review.
  - A detailed description of the proposed method and validation procedure, including evaluation metrics.
  - Planned algorithm hyperparameters and the strategy for selecting or tuning them.
  - Datasets and preprocessing procedures.
    - For tabular data, use at least a dozen regression datasets and a dozen classification datasets.
  - The approach to aggregating results.
  - Planned statistical tests.
- Any deviations from the plan must be justified.

Report

The report should include:

- Abstract, placed before the table of contents.
- Research problem: a self-contained description understandable to readers unfamiliar with the assignment.
- Theoretical background and literature review.
- Experimental setup: methods, datasets, preprocessing, hyperparameters, and evaluation procedures.
- Results and statistical analysis, presented clearly using appropriate tables and figures.
- Conclusions and discussion: key findings, possible explanations for successes and failures, limitations, and directions for future research.
- Instructions for running the code and reproducing the results.

Report: Additional remarks

- Document every experiment: experiments not described in the report will be treated as not conducted.
- Use a formal academic structure: include a table of contents, references, numbered figures and tables with captions, and references to them in the text.
- Interpret the results: explain the findings and observed patterns, and support your conclusions with evidence.
- Repeat experiments where feasible to assess variability and support reliable statistical analysis. Repetition alone does not guarantee statistical significance.
- Report both the mean and standard deviation: more stable performance may be preferable even when average performance is slightly worse.

Potential publication venues

- IJCAI and ICML.
- IJCNN and ICCS.
- A peer-reviewed journal.
- PP-RAI, ICAISC.

Intuition (1)

Poprawna klasa	Model 1	Model 2	Model 3	Zespół
A	A	A	A	A
A	A	A	A	A
B	A	A	A	A
A	B	B	B	B
B	A	A	A	A
B	B	B	B	B
B	B	B	B	B
A	B	B	B	B
B	B	B	B	B
A	A	A	A	A
Accuracy	60%	60%	60%	60%

Intuition (2)

Poprawna klasa	Model 1	Model 2	Model 3	Zespół
A	A	A	A	A
A	A	A	A	A
B	A	A	A	A
A	B	B	B	B
B	B	B	A	B
B	A	B	B	B
B	B	A	B	B
A	B	B	B	B
B	B	B	B	B
A	A	A	A	A
Accuracy	60%	60%	60%	70%

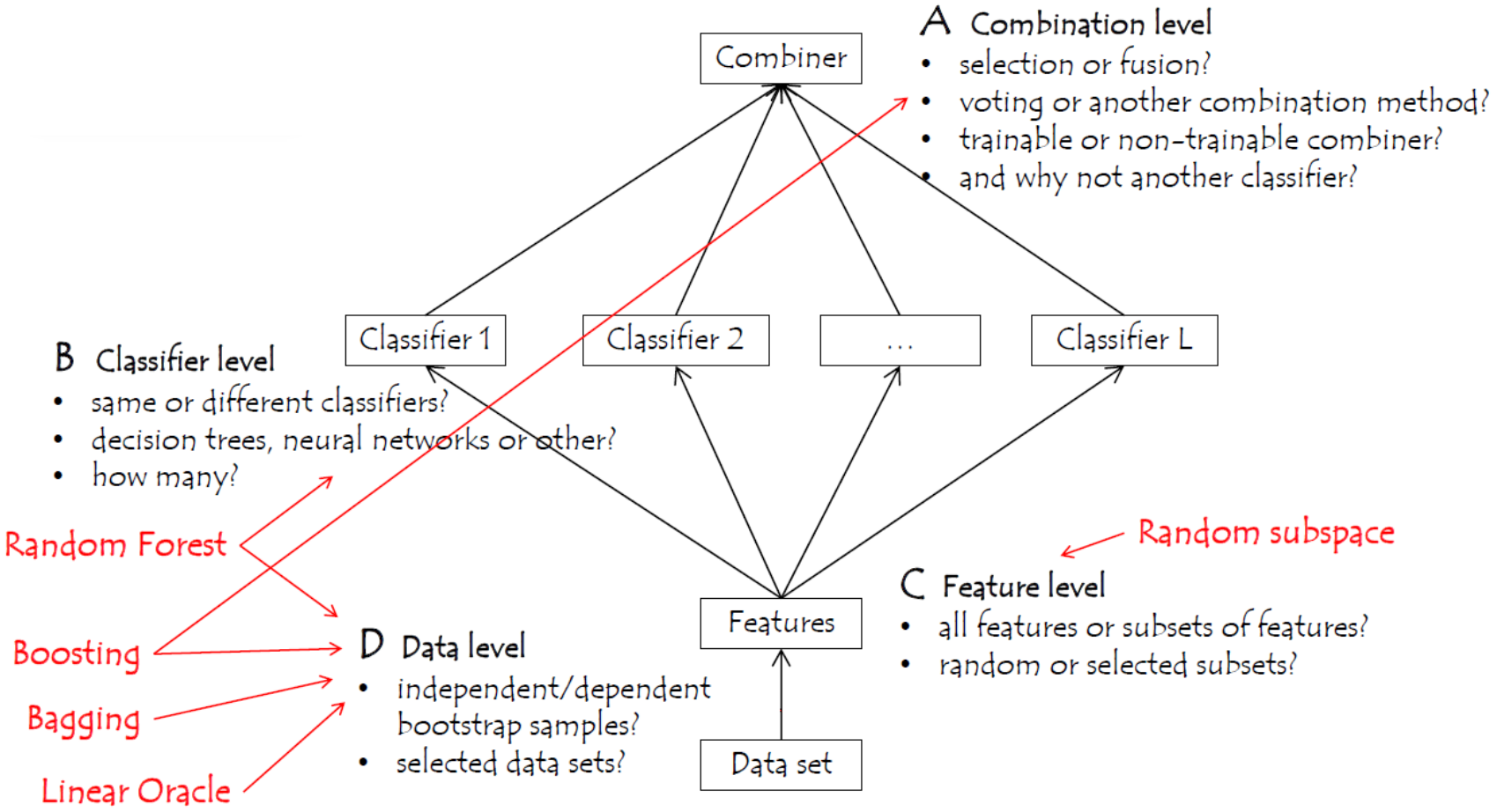
Intuition (3)

Poprawna klasa	Model 1	Model 2	Model 3	Zespół
A	A	B	A	A
A	A	A	B	A
B	A	B	B	B
A	B	B	B	B
B	B	B	A	B
B	A	B	B	B
B	B	A	B	B
A	B	A	A	A
B	B	B	A	B
A	A	A	A	A
Accuracy	60%	70%	60%	90%

Motivation

- Proven competition success: many winning Kaggle solutions combine predictions from multiple models.
  - „ Last step: ensemble it”
- Strong benchmark performance: ensembles have powered winning solutions on benchmarks such as ImageNet and COCO.
- A simple starting point: train multiple instances of the same model with different random seeds or training samples, then combine their predictions through averaging or voting.
- Better accuracy and stability: combining models with complementary errors can improve generalization and reduce prediction variance.

Ensemble learning – different approaches



[1]

Project topics

- Any proposed topic with research potential identified through a literature review.
- A link to example project topics is available on the LeON platform.
- Remark 1: Topics related to random forests can also be explored in the context of other tree-based ensemble models, such as gradient boosting or extremely randomized trees (Extra Trees).
- Remark 2: Random forests and gradient boosting are the most popular (but not the only!) implementations of bagging and boosting, respectively.

Sources

1. Kuncheva, L. I. (2016, February 24–26). *Getting lost in the wealth of classifier ensembles?* [Keynote presentation]. 5th International Conference on Pattern Recognition Applications and Methods (ICPRAM 2016), Rome, Italy.